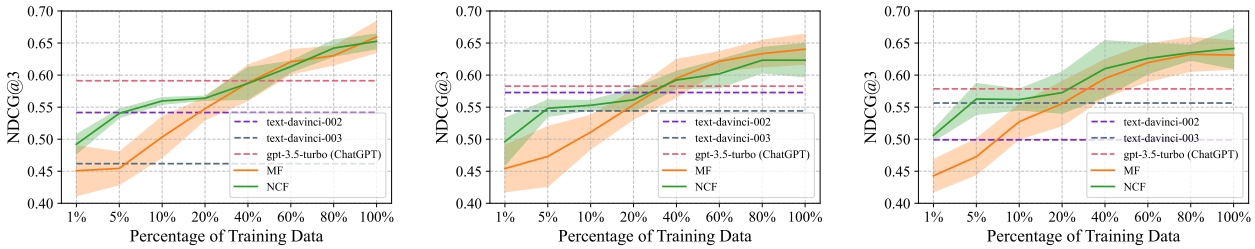
Table 3: Performance of different LLMs with zero-shot and few-shot examples on Movie dataset. Bold indicates the best result for each row and '_' indicates the best result for each wise of each LLM.

Model	Metric	random	pop	point-wise		pair-wise		list-wise	
				zero-shot	few-shot	zero-shot	few-shot	zero-shot	few-shot
text-davinci-002	NDCG@3	0.4264	0.4761	0.5168	0.5416	0.5253	**0.5728**	0.4544	0.4990
	MRR@3	0.3667	0.4103	0.4519	0.4824	0.4643	**0.5071**	0.3950	0.4363
text-davinci-003	NDCG@3	0.4264	0.4761	0.4674	0.4618	0.5249	**0.5441**	0.5062	0.5564
	MRR@3	0.3667	0.4103	0.4092	0.3998	0.4633	**0.4763**	0.4450	0.4950
gpt-3.5-turbo (ChatGPT)	NDCG@3	0.4264	0.4761	0.5413	**0.5912**	0.5833	0.5827	N/A	0.5785
	MRR@3	0.3667	0.4103	0.4742	**0.5260**	0.5243	0.5162		0.5167



(a) point-wise (b) pair-wise (c) list-wise

Figure 2: Comparison with collaborative filtering models in terms of different percentages of training data on Movie dataset. The shaded area indicates the 95% confidence intervals of t-distribution under 5 different experiments with random seeds.

Table 2. Note that pair-wise ranking tends to be better than point-wise ranking in most cases (11 out of 12), although it requires more inference cost due to the need for pair-wise comparisons. We will delve deeper into the cost analysis in **RQ3**.

All LLMs-based recommenders outperform the popularity recommendantion policy in recommending movies, books, and music, but they underperform in the news domain. This phenomenon could be explained by the fact that news recommendation relies more on popularity, while other domains are more personalized. The speed of news delivery is another possible factor. Due to the time-sensitive and rapidly changing nature of news recommendation, there is often insufficient interaction data available for each news in the LLMs training corpus. Conversely, in the other three domains, the item descriptions and interaction data are more abundant, making LLMs works better on them. Overall, this observation suggests that while off-the-shelf LLMs-based recommenders can be effective in many domains, they may not be suitable for some domain and may require further exploration.

We also conduct experiments using zero-shot prompts (i.e., without examples). However, with the original zero-shot prompt, we find more than 50% of cases were invalid and challenging to evaluate. To address this, we utilize logit_bias[5] to control the output tokens. Due to the page limitation, we provide the detailed results in the link[6].

[5]https://platform.openai.com/docs/api-reference/completions/create#completions/create-logit_bias
[6]https://github.com/rainym00d/LLM4RS/blob/main/assets/Supplementary_Material.pdf

Overall, the results highlight the potential of LLMs as recommendation systems, as they outperform random and popularity-based policies in the zero-shot setting. Furthermore, as expected, LLMs under few-shot settings outperform those under zero-shot settings in most cases, demonstrating the effectiveness of few-shot prompts in-context learning.

4.3 RQ2: Comparison with Collaborative Filtering Models

Given that the LLMs used in the previous experiments were not trained on recommendation data, we investigate the amount of training data required for traditional recommendation models to achieve performance comparable to or better than LLMs. Specifically, we chose the most representative traditional recommendation models, Matrix Factorization (MF) [17] as well as Neural Collaborative Filtering (NCF) [12], and evaluated their performance after training on varying proportions of data. For a fair comparison, we carefully tune the parameters of MF and NCF. We then compared their performance to that of LLMs. All experiments are conducted 5 times on the Movie dataset, and the averaged results and their 95% confidence intervals of $NDCG@3$ were illustrated in Figure 2. As expected, the performance of MF and NCF improves with increasing amounts of training data. Also, we can observe that off-the-shelf LLM-based models outperform MF and NCF when there are only a few training data available, i.e., less than 40% for ChatGPT with all three ranking capabilities. Note that LLM-based recommendation